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January 22, 1971
SUPERSEDING
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January 29, 1965

FEDERAL SPECIFICATION

STOOL, STEP, CASTER MOUNTED, SAFETY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers minimum general requirements for a caster mounted, step stool of commercial design.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified, (see 6.2), before production is commenced, a sample of the finished commodity and each component part shall be submitted or made ready for the contracting officer, or his authorized representatives to examine and test to determine compliance with this specification. Approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of responsibility from compliance with all other applicable provisions of this specification. Production units shall not vary from the approved preproduction sample in design or construction without written approval of the contracting officer.

3.2 Material.

3.2.1 Steel. All sheet steel parts shall be cut, stamped or die formed from FS 1010-1020 cold rolled steel. The steel shall be smooth, free of scale, pits, wrinkles, laps, crimps, or buckles affecting strength or appearance. Steel tubing, when used, shall be of round or square section, either seamless or continuously seam welded.

3.2.2 Plastic for body. Plastic, when used for the body of the stool, shall be impact type, molded polystyrene copolymer. It shall be Styron formulation 475, or any other impact type molded polystyrene copolymer formulation with similar typical property values. The color of the plastic shall be gray.

3.2.3 Tread facing material. Except as permitted herein the tread facing material for both top and bottom steps shall be a natural or synthetic molded rubber compound or molded vinyl plastic, either of which shall have a Shore Durometer hardness of 65 to 75. Alternatively, the tread facing material for the bottom step only may be a molded styrene and butadine copolymer with resilient and elastic properties. All tread facing materials shall be corrugated or ribbed, shall be black, and shall be at least 7/64 inch thick.

3.2.4 Hardware.

3.2.4.1 Casters. Casters shall be a standard commercial product with wheels of a hard rubber, hard phenolic, or molded hard polystyrene compound. Wheel diameter shall be not less than 1-1/4 inches. Casters shall swivel freely.

3.2.4.2 Fastening devices. Bolts, nuts, screws, washers and similar fastening devices shall be made to resist rust by electo-galvanizing or by zinc, chromium or cadmium plating in accordance with the manufacturer's commercial practice.

3.2.5 Finishing materials.

3.2.5.1 Primer. The primer, when used, shall be compatible with the finishing enamel used.

3.2.5.2 Enamel. Enamel for the finishing of steel stools shall be the baking type and shall be the manufacturer's commercial product. Color of the finish shall be gray.

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3.3 Design and construction.

3.3.1 Design. The step stool shall consist of a caster mounted frame with two step levels arranged one above the other with the upper level of smaller area than the total step area on the lower level. The stool shall have an automatic safety feature actuated by a relatively slight weight applied to the surface of the steps. It shall roll freely on casters when unweighted and shall become stationary as soon as a weight of not more than 20 pounds is placed on either step. Step areas shall be accessible from at least two directions.

3.3.2 Construction, general. The body or frame of the stool shall be constructed of steel in accordance with 3.2.1 or plastic in accordance with 3.2.2. The stool may be either sectional or one piece. When constructed of steel, the components shall be welded, bolted or riveted together except steel sections may be attached by means of interlocking. Plastic stools shall be of molded construction and plastic sections shall be attached by means of interlocking. Formed steel parts shall have true and even curves without flattening, crushing or crimping that affect the strength or appearance of the parts. There shall be no sharp corners or protrusions projecting from any surface of the stool. Where bolt heads are exposed, only flat head bolts with heads countersunk or oval head bolts shall be used. If of tubular construction, the ends of the tubing shall be closed. Sectional stools may be furnished knocked-down to the extent that the sections need not be attached, provided all necessary attaching devices and complete assembly instructions are included with each knocked-down stool.

3.3.3 Dimensions.

3.3.3.1 Height. The overall height of the stool, when unweighted shall be not less than 14 inches nor more than 16 inches.

3.3.3.2 Step areas. The surface area of the top step shall be not less than 85 square inches and the total surface area of the bottom steps shall be not less than 125 square inches.

3.3.3.3 Distance from floor to first step and between steps. The distance from the floor to the first step and between the first steps and the top step shall not exceed 8-1/4 inches. The overall height specified in 3.3.3.1 shall be maintained.

3.3.4 Tread. The step tread facing shall be one of the materials specified in 3.2.3. Only rubber or vinyl plastic shall be used on the top step. The facing material shall extend to within 1/2 inch of the outside edge of each step and shall cover a sufficient area of each step to provide a firm footing for both feet of the user simultaneously. It shall be affixed throughout the area of contact with the step surface with an adhesive that will assure proper bondage.

3.3.5 Contact surfaces. All points of the stool that contact the floor when the stool is supporting weight shall be covered with an elastomer which will not mar floors and is resistant to sliding.

3.3.6 Performance characteristics.

3.3.6.1 Stability. When tested in accordance with 4.5.1.1.1 and 4.5.1.1.2, the stool shall not wobble, roll or overturn.

3.3.6.2 Strength. The stool shall have the strength to withstand the service test specified in 4.5.1.2 and the static load test specified in 4.5.1.3.

3.4 Pretreatment and finishing of metal surfaces.

3.4.1 Pretreatment. All exterior and interior unplated metal surfaces shall be treated for painting in accordance with any method of TT-C-490.

3.4.2 Priming. Unless type I or type II of TT-C-490 is used, all exterior and interior surfaces treated as specified in 3.4.1 shall be coated with primer as specified in 3.2.5.1.

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3.4.3 Enameling. All primed surfaces shall be coated with the enamel specified in 3.2.5.2 and baked in accordance with the enamel supplier's directions. Total dry film thickness of enamel and primer shall be not less than 1.0 mil on exterior surfaces and not less than 0.6 mil on interior surfaces. Primer may be omitted from any surface prepared in accordance with Type I or Type II of TT-C-490 provided the dry enamel thicknesses are the 1.0 mil and the 0.6 mil specified above. The finish shall level out to provide smooth, uniform, exposed surfaces without runs, wrinkles, grit, areas of thin film or no film, or separation of color. Finish panels shall withstand the test specified in 4.5.2.

3.5 Identification marking. Unless otherwise specified, each stool shall be permanently and inconspicuously marked in a plain and legible manner with the specification number, Federal Stock Number, Contract number, month and year of manufacture and the manufacturer's name or trade mark, so that the source of supply can be readily determined. When specified (see 6.2) stools shall be marked for identification in accordance with MIL-STD-130.

3.6 Workmanship. The stool shall be constructed in a manner to insure a well balanced, substantial and easy operating item, without sharp or hazardous features or surfaces, or components that will work loose during use. The forming of component parts, welding, reinforcing, bracing, curves of bent tubing, when used, and accuracy of moving parts shall be executed so that the finished items will meet the quality level established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. Examination and testing of the preproduction samples (when required), shall be on a completely constructed item, before regular production is started, for all provisions of this specification applicable to end product examination and tests.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with MIL-STD-105 except where otherwise specified herein.

4.3.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards, or if none, in accordance with this specification.

4.3.2 End item inspection.

4.3.2.1 Visual examination. The stools shall be examined for the defects listed in table I. The inspection level and acceptable quality level shall be as specified in 4.4. The lot shall be all stools offered for inspection at one time. The lot size shall be expressed in units of stools for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit for this examination shall be one completely assembled stool.

TABLE I. Classification of end item defects

Examine	Defect	Classification Major	Classification Minor
Enamel finish	Poor adhesion, i.e. blistered, checked, peeling; not dry to touch, color separation, stain, not smooth and uniform, runs, sags, foreign matter in coating, area of thin coating or abrasion.		X
	Thickness not as specified.		X
	Scratch through to base metal, bare spots, runs under coating.	X	

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TABLE I. Classification of end item defects (con't.)

Examine	Defect	Classification Major	Classification Minor
Metallic coating	Wrong kind.		X
	Poor adhesion, i.e. blistered, peeling, flaky, stained or discolored.		X
Workmanship and construction	Any part missing, including fastening devices.	X	
	Malformed or mislocated parts affecting serviceability; poor fit of component parts; broken, punctured or cracked parts; sharp edge or projection that may cause injury or tear clothing.	X	
	Malformed, mislocated, damaged, buckled or warped parts affecting appearance; any dent on exposed surface affecting appearance.		X
	Not material specified:	X	
	Ends of tubes not closed.		X
	Welds missing or not sound (broken, cracked, burnt through, poor fusion, etc.)	X	
Identification marking	Missing, not permanent, or is illegible, incomplete or incorrect.		X

4.3.2.2 Dimensional examination. Inspection will be made of stools for compliance with dimensions specified. Any dimension not within the tolerance specified will be classified as a defect. The inspection level and acceptable quality level shall be as specified in 4.4.

4.3.3 Examination of preparation for delivery. An examination will be made to determine compliance with packaging, packing and marking provisions as required by section 5 of this specification. Defects will be scored in accordance with table II. The sample unit shall be one shipping container. The lot size shall be the number of containers in the inspection lot. The inspection level and acceptable quality level shall be as specified in 4.4.

TABLE II. Preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Any nonconforming components; components missing, damaged, or any other defect affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate sealing; bulging or distortion of container.

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4.4 Inspection levels and AQL's. The inspection levels and the acceptable quality levels, (AQL's), expressed in defects per hundred units, shall be as follows:

For examination in	Inspection levels	AQL's Major	AQL's Total
Table I	II	4.0	10.0
4.3.2.2 (1 class)	S-4		4.0
Table II (1 class)	II		4.0

4.5 Tests.

4.5.1 Performance tests. The tests in 4.5.1.1.1 through 4.5.1.3 shall be performed on finished stools to determine compliance with 3.3.6. Inspection level S-1 shall be used for selection of the number of stools to be tested from any lot. Failure to pass any test shall be cause for rejection of the lot represented by the sample stool under test. Tests in 4.5.1.2 and 4.5.1.3 shall be performed on the same stool. One stool may be used for all tests.

4.5.1.1 Tests for stability. The test described in 4.5.1.1.1 determines the degree to which the stool becomes immobile and free from wobble when a relatively slight weight is applied. The test described in 4.5.1.1.2 determines the degree to which the design of the stool precludes tipping in use.

4.5.1.1.1 Test for immobility and freedom from wobble. The stool shall be placed on a flat level surface with the casters in rolling position. A weight of 20 pounds shall be applied to the central area of the top step. Failure of the stool to become immobile shall constitute a failure of the test. If the stool has not failed at this point, and with the weight remaining on it, vertical pressure shall be applied to the top step at several points near its perimeter. Indication of wobble shall constitute non-compliance with 3.3.6.1. The stool shall be further tested by removing the 20 pound weight from the top step and applying it gradually to the lower step at each accessible direction of the stool, within two inches of the step's perimeter. Either of the following shall constitute failure of this part of the test:

(a) Rolling of the stool during application of the weight.
(b) The failure of at least part of the bottom to contact the floor (or test surface).

4.5.1.1.2 Test for balance. A weight of 150 pounds shall be applied (1) to the top step of the stool within two inches of its perimeter (2) opposite the first location, within 2 inches of the perimeter of the top step and (3) to the lower step at each of its accessible directions and within 2 inches of the step's perimeter. Only one 150 pound weight shall be on the stool at any time during performance of this test. The overturning of the stool while the weight is at any of the locations shall constitute failure of the test.

4.5.1.2 Service test. A 3-1/2 inch wide by 12-inch long plywood strip shall be placed across the central area of the top step. An equally distributed load of 350 pounds shall be positioned on the strip and cycled 500 times at the rate of 12 strokes per minute. One cycle shall consist of lifting the weight so that the stool is elevated full height and returning the weight to full load. After 500 cycles the weight shall be reduced to 250 pounds and an additional 1500 cycles completed. At the completion of 2000 cycles there shall be no fractures, cracks, permanent deformation or other structural failure.

4.5.1.3 Static load test. A 3-1/2 inch-wide by 12-inch long plywood strip shall be placed across the top of the stool. An equally distributed load of 500 pounds shall be positioned on the strip and allowed to remain for 30 minutes. After removal of the load there shall be no fractures, permanent deformation or other structural failure.

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4.5.2 Finish tests. The finish tests shall be performed on 20 gage steel sample panels prepared in accordance with 3.4.

4.5.2.1 Flexibility. The dried film shall show no evidence of cracking or flaking under seven power magnification after a panel has been bent through 180 degrees over a 1/8-inch diameter rod.

4.5.2.2 Hardness. The dried film shall withstand the firm stroke of a 2H pencil held at a 45° angle and pushed across the film surface without evidence of marring when viewed at an oblique angle in a strong light.

4.5.2.3 Adhesion. The dried film shall not be removed from the panel when the latter has been scored with a razor blade through the film to the base metal in such a manner as to produce a grid of 1/8-inch squares, and a one inch wide piece of cellophane tape is applied firmly to the grid surface and then quickly pulled from the surface.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each stool shall be packaged in a close-fitting box conforming to PPP-B-636, W5c. Scored and folded, die-cut, or built-up fiberboard pads shall be used to cushion the stool in the box. Each box shall be closed, waterproof sealed, and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.1.2 Level B. Each stool shall be packaged as specified in 5.1.1, except the box shall be class-domestic and waterproof sealing and reinforcing with tape shall not be required. Closure shall be in accordance with the appendix to the box specification.

5.1.3 Level C. Each stool shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The stools, packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601; overseas type, PPP-B-621, class 2 or to PPP-B-636 V3c. The boxes shall be closed and strapped in accordance with the appendix to the applicable specification. The gross weight of the wood boxes shall not exceed approximately 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Unless otherwise specified (see 6.2) the stools packaged as specified in 5.1.2 shall not require additional packing for shipment. When specified (see 6.2), the stools shall be packed in a box conforming to PPP-B-636, class-domestic. Closure shall be in accordance with the appendix to the box specification.

5.2.3 Level C. The stools shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES.

6.1 Intended use. The step stool covered by this specification is intended primarily to afford safe access to areas not within normal arm's reach. It can be used to provide seating when work is being performed in lower drawers or on lower shelves, however; when this requirement alone exists, the Type I, size 1 stool of Federal Specification AA-S-700 should be used.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) If preproduction sample requirement is waived (3.1).
(c) If special marking is required (3.5 and 5.3).
(d) Level of packaging and packing required (5.1 and 5.2).
(e) If additional packing required for level B shipment (5.2.2).

Preparing activity:

GSA-FSS

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